

CHAPTER 8

GASES

By the end of this chapter you will have covered the following material.

Science Understanding

- The behaviour of gases, including the qualitative relationships between pressure,

temperature and volume, can be explained using kinetic theory ([ACSCH060](#))



8.1 Introducing gases

We live at the bottom of a layer of gases called the atmosphere and we breathe air from the atmosphere into our bodies every few seconds. Life could not exist without the oxygen and carbon dioxide that make up part of our atmosphere. However, gases often go unnoticed except when they are at their most destructive in cyclones and tornadoes. Every day, we use gases for many purposes, including adding fizz to soft drinks, liquefied petroleum gas (LPG) to fuel motor vehicles and natural gas for cooking and heating.

Figure C8.1 ►
The atmosphere as seen from space



istockphoto/Baoshan Zhang

Figure C8.2 ►
Some products that use components of air.



Debra Smith



USING AIR

Watch this video showing how gases are removed from the air, used in food packaging and transported.

Air

Air is a mixture of many substances. The main four are the gases nitrogen, oxygen, argon and carbon dioxide. The amounts of these are remarkably constant throughout the atmosphere. Air also contains water vapour, but its percentage varies considerably from close to 0% in dry desert air to 5–6% in air above a tropical rainforest. Scientists usually refer to the composition of dry air (Table C8.1) on page 343.

Table C8.1 Major and minor components of dry air

Major constituent	Amount (% v/v)	Minor constituent	Amount (ppm)
Nitrogen	78.08	Neon	18
Oxygen	20.95	Helium	5
Argon	0.93	Methane	1.5
Carbon dioxide	0.035	Krypton	1
		Hydrogen	0.5
		Dinitrogen oxide	0.2
		Carbon monoxide	0.1
		Ozone	0.02
		Nitrogen oxide and nitrogen dioxide	<0.01
		Ammonia	<0.01
		Sulfur dioxide and dihydrogen sulfide	<0.002

The concentrations of some substances are measured in parts per million (ppm) by volume. This measure is often used when concentrations involved are very small; for example, 350 ppm corresponds to a percentage concentration of 0.0350%. Parts per million can also be used for weight/weight (w/w) comparisons.

In Table C8.1 the composition of the air is based on volume, so 100 L of air contains approximately 78 L nitrogen (N_2), 21 L oxygen (O_2) and 1 L argon (Ar). An alternative way to represent composition is in terms of relative amounts of particles or moles, but because the volume of a gas sample varies with temperature and pressure, if gas volumes are to be compared, then they must be measured at the same temperature and pressure

Gas pressure

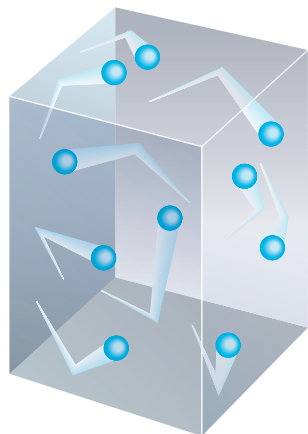
Gases consist of particles in constant random motion and the gas particles are a long way apart. When the gas particles collide with something, they exert a **pressure**. Gas pressure is a measure of the force per unit area. The International System of Units (SI) unit of gas pressure is the **pascal (Pa)** and is equivalent to a force of 1 newton per square metre.



◀**Figure C8.3** Hot air ballooning is possible because the hot air inside the balloon is less dense than the outside air, allowing the balloon to rise.

Figure C8.4 ►

Pressure is exerted by gas particles when they collide with the walls of a container.



EFFECT OF AIR PRESSURE

Watch this video, which demonstrates the effect of air pressure.

The gases in the atmosphere exert pressure on every exposed surface. The pressure due to the atmosphere is called **atmospheric pressure**. This pressure varies from place to place and time to time. At sea level, atmospheric pressure is like having a kilogram tub of margarine pushing on every square centimetre of every part of the surface of your body all the time but we don't usually notice it is even there.

In order to make communication easier, scientists agreed on a standard of pressure to represent the average air pressure at sea level. This unit of standard atmospheric pressure is 1.01325×10^5 Pa or 101.325 kPa (kilopascals) at sea level. Another common unit for pressure is the atmosphere, which is defined as:

$$1 \text{ atmosphere (atm)} = 101.3 \text{ kPa}$$

Atmospheric pressure varies with height above sea level, temperature and weather conditions, so it is not always 1.00 atm. Meteorologists generally mention high- and low-pressure systems in weather reports and use the hectopascal (1 ha = 100 Pa) or millibar (100 Pa) when discussing pressure. Chemists and physicists generally use the units pascal (or kilopascal) and atmosphere (Table C8.2).

Table C8.2 Commonly used units of gas pressure

Name of unit	Symbol	Comparison
Pascal	Pa	Standard unit, 1 newton per square metre
Kilopascal	kPa	1000 Pa
Atmosphere	atm	101.3 kPa
Hectopascal	hPa	100 Pa
Millimetres of mercury	mmHg	760 mmHg = 1 atm

WOW

Barometer

Italian scientist Evangelista Torricelli is credited with inventing the mercury barometer in 1643 to measure atmospheric pressure. He created a 1-metre-long tube that was sealed at one end, filled it with mercury and set it vertically in a basin of mercury. The column of mercury in the tube fell to a height of 76 cm and this height was originally used as a standard measure of atmospheric pressure. The obsolete unit of pressure, the torr, which equals 1 mmHg, is named after him.

Figure C8.5 ►

A barometer is used to measure atmospheric pressure.

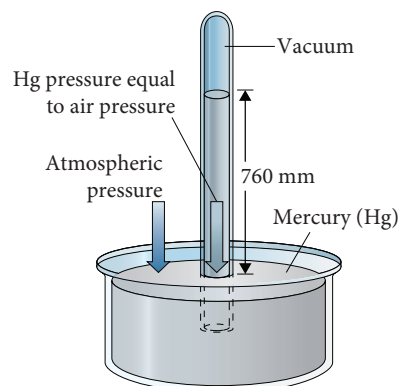


Figure C8.6 shows how to convert between different units of pressure.

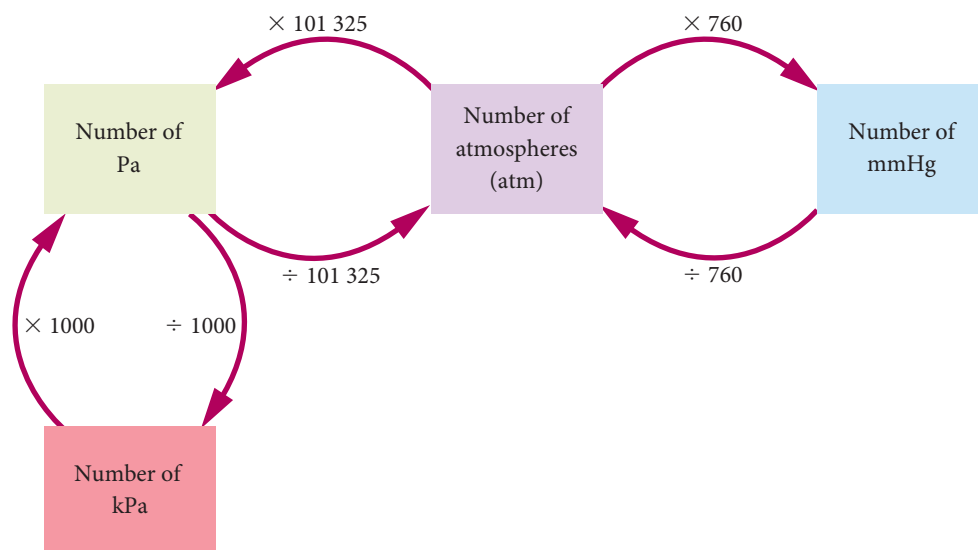


Figure C8.6
Converting pressure units



PRESSURE UNITS DEFINED

Watch this video, which describes different units and explains how to convert between units.

QUESTION SET 8.1

Remembering

- 1 Explain what causes gas pressure.
- 2 Name two standard units of pressure.

Understanding

- 3 Use Table C8.1 to answer the following.
 - a Draw a bar graph to show the composition of the main gases of the atmosphere.
 - b How much argon is in a 100 L of air?
- 4 Explain what causes air pressure to vary.
- 5 Convert the following pressures to atmospheres.
 - a 75.5 kPa
 - b 6.83 Pa
 - c 652 mmHg
- 6 Convert the following pressures to kPa.
 - a 1.03 atm
 - b 0.45 atm
 - c 820 mmHg

Applying

- 7 Decide whether the following objects would work if there were no atmosphere. Explain why or why not.
 - a Vacuum cleaner
 - b Siphon
 - c Match
 - d Magnet



To see how low pressure and amount of oxygen affect us, refer to Context 4, 'Making reactions work for us', page 92.

8.2 Kinetic theory of gases

The pressure exerted by gases is one of the many similar physical properties they exhibit. Experimental observations by chemists that different gases behave in a similar manner led to the development of mathematical relationships linking volume, mass, pressure and temperature. These relationships are based on the notion of an **ideal gas**; that is, one that obeys the relationships perfectly. While no real gas behaves in a completely ideal manner, most gases are considered to being close enough to ideal under normal conditions that the relationships can be used.

The **kinetic theory of gases** explains much of the physical behaviour of gases. The term 'kinetic' comes from the Greek *kinetikos*, which means 'moving', so many of the properties of gases are related to their motion.

The kinetic theory of gases proposes that:

- 1 Gases consist of molecules (except the noble gases, which consist of atoms) that move in continual random straight-line motion.
- 2 The average distance between gas molecules is very large compared to the size of the molecule.
- 3 Intermolecular forces between molecules are negligible.
- 4 All collisions of gas molecules are perfectly **elastic collisions**, which means there is no net energy loss during these collisions.
- 5 Pressure is due to collisions of the molecules with the walls of the container.
- 6 Temperature is a measure of the average kinetic energy of the molecules.

The kinetic theory of gases is used to explain the physical behaviour of gases.

It is important to remember that this theory uses the model of an ideal gas. Gases, such as hydrogen and helium, which have small, light particles come close to 'ideal' under conditions of low pressure and high temperature. Chemists apply kinetic theory to real gases under normal conditions.

Considering temperature

The kinetic theory states that temperature is a measure of average kinetic energy. Although the molecules of a sample of gas have the same average kinetic energy, the individual molecules move at various speeds. Some move faster and some move slower than the average. Collisions between particles change individual speeds.

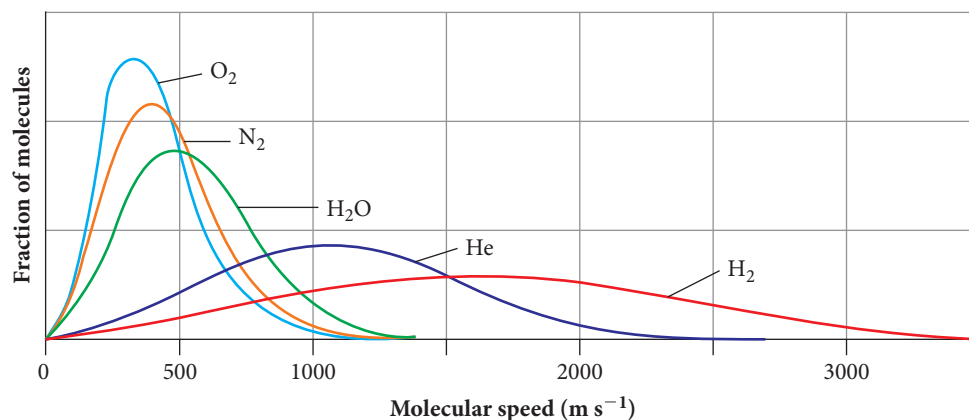
However, at a given temperature, molecules of all gases, no matter what size, shape or mass, have the same average kinetic energy. Kinetic energy is given by the formula:

$$\text{Kinetic energy} = \frac{1}{2}mv^2$$

where m is the mass and v is the speed.

Because kinetic energy is determined by both mass and speed, heavier gases will have lower speeds than lighter gases at the same temperature, as shown in Figure C8.7.

Figure C8.7 ►
Distribution of speeds
for different gases at the
same temperature



As the temperature increases, the kinetic energy of the particles increases, and vice versa. Although the gas particles have a range of speeds at any given temperature, the average speed of the particles of a gas increases as temperature increases. Figure C8.8 shows how.

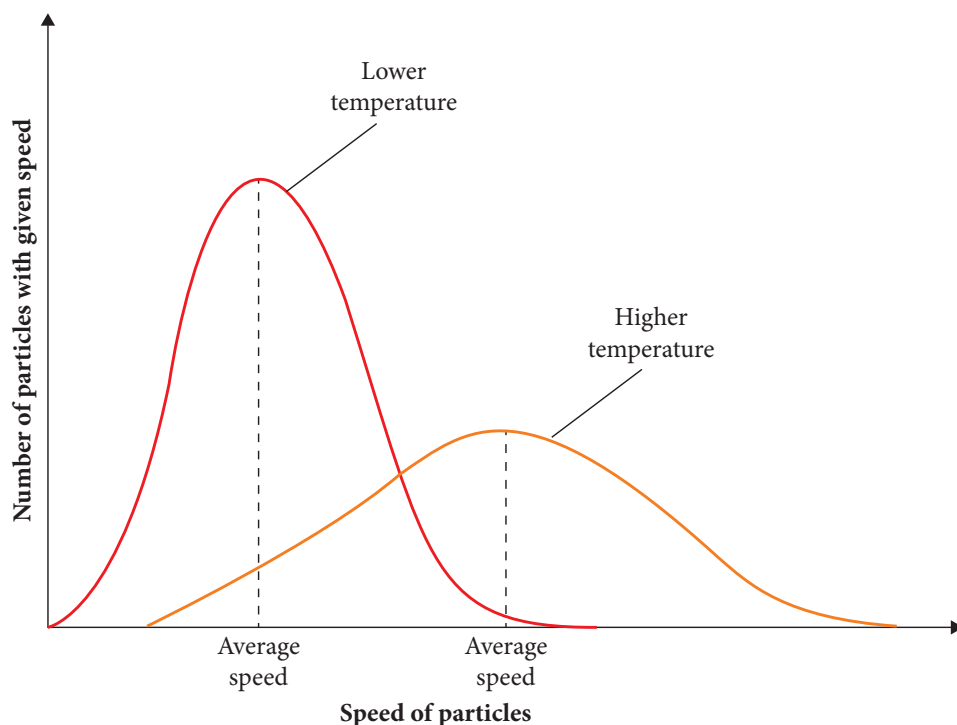


Figure C8.8
Average speed of particles at two different temperatures

Therefore, decreasing temperature will result in the particles slowing down. Theoretically, at low enough temperatures, the particles will eventually stop moving. The theoretical lowest temperature possible is known as **absolute zero** (0 K), which is equivalent to -273.15°C .

Absolute (kelvin) temperature scale

The **absolute temperature** is a measure of the average kinetic energy of its molecules. It was developed by British scientist William Thomson, who was also known as Lord Kelvin. He used experimental data and concluded from relationships between volume and temperature that there had to be a lower limit to temperature and that this limit was -273°C .

Lord Kelvin set up a new temperature scale and made the lowest number on the scale 0, which is also called absolute zero. The scale has the same sized units as the Celsius scale. Because he was the first person to recognise the importance of absolute zero, the unit on the scale is called the kelvin (K). The units do not use the degree sign. Figure C8.9 shows how to convert between Celsius and absolute temperature scales.

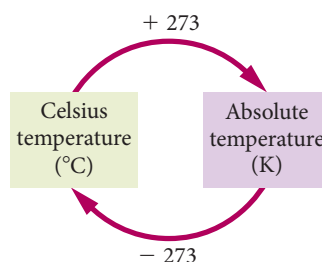


Figure C8.9
Converting temperature units

WOW

A new state of matter

In 1924–25 Satyendra Bose and Albert Einstein predicted that a new state of matter would appear as a gas approached absolute zero. This state was called the Bose–Einstein condensate. Seventy years later, in 1995, Eric Cornell and Carl Wieman produced the first Bose–Einstein condensate using a gas of rubidium atoms cooled to 1.7×10^{-7} K.



ABSOLUTE ZERO

Watch this interview of scientists who almost achieved absolute zero.

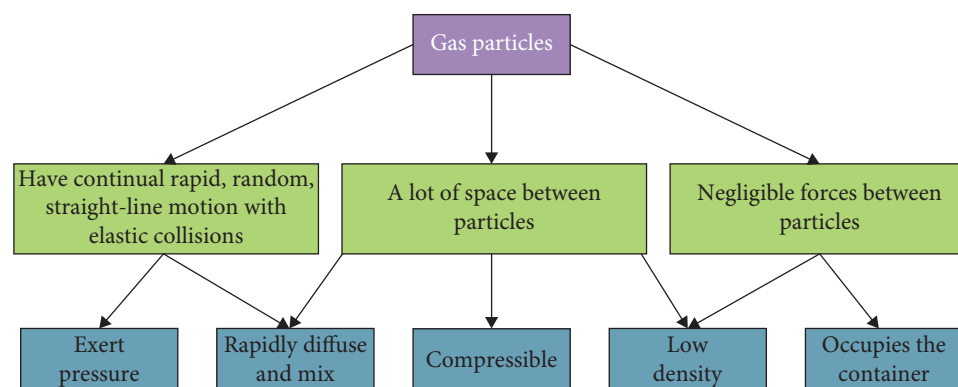


Applying the kinetic theory

The kinetic molecular theory was developed using the findings of many different scientists. It can be used to explain many properties of gases.

- Gases **diffuse**, or spread, because gas particles have rapid, random motion and a lot of empty space between them so they quickly spread and mix.
- Gases can be easily compressed because there is a lot of empty space between particles.
- Gases spread to fill a container because there are negligible forces between particles.
- Gases exert pressure because the particles move rapidly and collide with the surface of any container or object.
- Gases have low density because there is a lot of empty space between them.

Figure C8.10 ►
The kinetic molecular theory of gases – a schematic diagram



QUESTION SET 8.2

Remembering

- 1 List the proposals of the kinetic theory.
- 2 What is meant by elastic collisions between particles?
- 3 What is absolute zero?

Understanding

- 4 Use the kinetic theory to explain why gases:
 - a can be compressed.
 - b can mix rapidly.
 - c have low density.
- 5 Explain why an increase in the temperature of a gas-filled balloon will cause it to burst.
- 6 What would happen to the volume of a gas-filled balloon if gas collisions were not elastic?
- 7 Explain why the particles of hydrogen gas move faster than those of oxygen gas at the same temperature when they have the same kinetic energy.
- 8 What is the difference between an ideal gas and a real gas?
- 9 Convert the following temperatures.
 - a 237°C to K
 - b -25°C to K
 - c 120 K to °C
 - d 345 K to °C

Applying

- 10 On a hot day, a metal can is half-filled with petrol and sealed. The next morning, after a cold night, it is observed that the can is partially crushed. Explain why this could happen.
- 11 Why do real gases behave more ideally at high temperature and low pressure?

12 At a temperature of 25°C, molecules of which of the following gases would move the fastest?

- A N_2
- B F_2
- C CO_2

Analysing

13 What is wrong with saying the gas bottle is half-full?

14 Suppose you have two containers, one filled with ammonia gas (NH_3) and the other filled with chlorine gas (Cl_2). If you were the same distance from both and they were opened at the same time, which one would you expect to smell first? Why?

8.3 Gas laws

You have already learnt that there are relationships between pressure, volume and temperature of a gas. When a balloon is heated, it gets bigger, thus maintaining a constant pressure. If a sealed container is exposed to extreme heat, such as that from a fire, the gas pressure can build up and the container may explode.

Standard conditions

In order to compare changes in conditions that involve gases, scientists use a set of standard conditions for temperature and pressure. These are defined in Table C8.2.

Table C8.2 Standard conditions for temperature and pressure

Standard conditions	Temperature (°C)	Pressure (kPa)
Standard temperature and pressure (STP)	0	100
Standard laboratory conditions (SLC)	25	100



Alamy/Tengku Mohd Yusof

Figure C8.11

A gas cylinder after an explosion

Boyle's law

More than 300 years ago, Robert Boyle was the first chemist to perform quantitative experiments on gases. By measuring the pressures and volumes of many different gases, he observed that, at a constant temperature, the volume of a given mass of gas is inversely proportional to the pressure. This is known as **Boyle's law**.

In mathematical terms, Boyle's law can be expressed as $V \propto 1/P$, which means the larger the applied pressure, the smaller the volume.

Another way of stating Boyle's law is $PV = k$, where k is a constant. For a particular sample of gas at a constant temperature, the product of the pressure and volume is a constant value.

Boyle's law states that at a constant temperature, the volume of a given mass of gas is inversely proportional to the applied pressure.



BOYLE'S LAW

View the NASA simulation of Boyle's law.

A useful alternative form of Boyle's law is:

$$P_1 V_1 = P_2 V_2$$

where P_1 is the initial pressure

V_1 is the initial volume

P_2 is the final pressure

V_2 is the final volume.

Note that any units for pressure and volume can be used, as long as they are consistent.

In Experiment 8.1, you can observe Boyle's law in action.

WORKED EXAMPLE 8.1

A sample of gas collected in a 242 mL container has a pressure of 87.6 kPa. What would the volume of this gas be at 101.3 kPa? (Assume temperature remains constant.)

Answer

The volume would be 209 mL.

Logic

1 Write the relationship:

$$P_1 V_1 = P_2 V_2$$

2 Determine the quantities:

$$P_1 = 87.6 \text{ kPa}, P_2 = 101.3 \text{ kPa}$$

$$V_1 = 242 \text{ mL}, V_2 = ?$$

3 Check the units are consistent.

4 Substitute in the formula:

$$87.6 \times 242 = 101.3 \times V_2$$

$$V_2 = \frac{87.6 \times 242}{101.3}$$

$$= 209 \text{ mL}$$

Try these yourself

Assume temperature is constant.

- a If 4.42 L of gas was collected at a pressure of 94.2 kPa, what volume would it occupy at 1 atm pressure?
- b A 210 mL sample of nitrogen gas at 1.0 atm is compressed until the final volume is 150 mL. What is the final pressure?
- c When the pressure is 522 kPa, a gas has a volume of 55 mL. What is its volume at atmospheric pressure?
- d A sample of hydrogen gas at 73.3 kPa in a 2.5 L flask is expanded into a total volume of 7.2 L. Calculate the final pressure.

EXPERIMENT 8.1

PRESSURE AND VOLUME

Cartesian divers are simple but fascinating toys that have been around for centuries. They consist of an object suspended near the top of a flexible, liquid-filled container. Cartesian divers are useful in demonstrating the relationship between pressure and volume.

Aim

To investigate pressure-volume relationships using a Cartesian diver

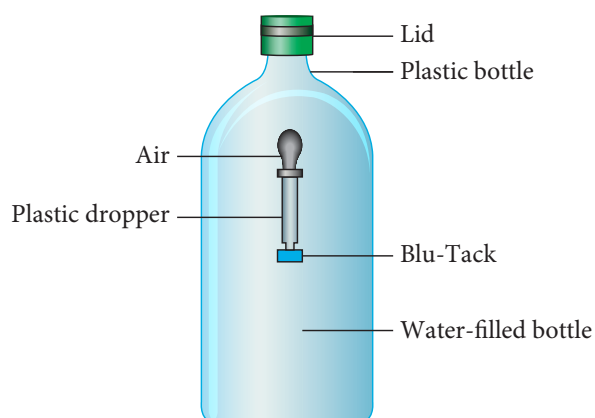


Materials

- 1 L clear, flexible screw-cap bottle with lid
- Clear plastic dropper
- Plasticine or Blu-Tack
- Water

Procedure

- 1 Totally fill the bottle with water.
- 2 Cut the plastic dropper about 1 cm below the bulb.
- 3 Half-fill the dropper bulb with water and weight it with some plasticine or Blu-Tack around the bottom, being careful not to block the hole.
- 4 Place the weighted dropper bulb in the bottle of water so it floats near the top. (You may need to adjust the amount of weight.)
- 5 Screw the top on the bottle and squeeze the container.



◀Figure C8.12 Experimental set-up for a Cartesian diver

Results

Describe what happens to the Cartesian diver when the bottle is squeezed and released.

Discussion

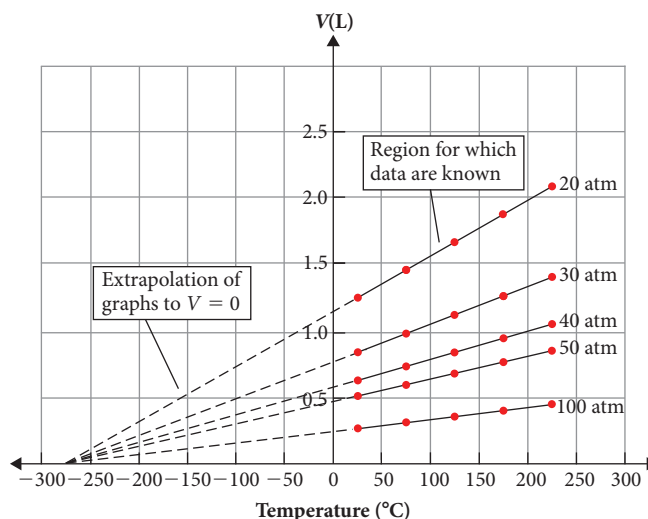
Explain what is happening in terms of Boyle's law and densities.

Charles' law

About 100 years after Boyle investigated the relationship between pressure and volume, French physicist Jacques Charles studied the relationship between temperature and volume using gases. The results of his many experiments led him to determine that for a fixed quantity of gas at a constant pressure, the volume increases linearly with temperature. This relationship is called

Charles' law.

When experimental temperature and volume data are plotted, a linear relationship emerges, as can be seen in Figure C8.13.



◀Figure C8.13 The graph of volume for 1 mole of hydrogen gas at different temperatures

Extrapolation of the data leads to a temperature of -273°C , which is absolute zero. So Charles's experiments led to the development of a new temperature scale. If the data is plotted using the absolute temperature scale, all lines will pass through the origin of the graph and therefore temperature (K) is directly proportional to volume. Therefore, in calculations involving gases, temperature must be given in kelvin.

Charles' law states that at constant pressure, the volume of a fixed quantity of gas is proportional to its absolute (kelvin) temperature.

In mathematical terms, Charles' law can be expressed as $V \propto T$ or:

$$\frac{V}{T} = \text{constant}$$

An alternative, more practical, form of this equation is:

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

where V_1 is the initial volume

T_1 is the initial temperature (K)

V_2 is the final volume

T_2 is the final temperature (K).

Note that any unit for volume can be used, as long as it is used consistently, but temperature must be in kelvin.

WORKED EXAMPLE 8.2

A 225 mL volume of gas is collected at 58°C . What volume would this sample occupy at 25°C ?

Answer

The volume would be 202 mL.

Logic

- 1 Write the relationship:

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

- 2 Determine the quantities and convert temperature to kelvin by adding 273:

$$V_1 = 225 \text{ mL}, V_2 = ?$$

$$T_1 = 58^{\circ}\text{C} = 331 \text{ K}, T_2 = 25^{\circ}\text{C} = 298 \text{ K}$$

- 3 Check the units are consistent.
- 4 Substitute in the formula:

$$\begin{aligned}\frac{225}{331} &= \frac{V_2}{298} \\ V_2 &= \frac{225 \times 298}{331} = 202 \text{ mL}\end{aligned}$$

Try these yourself

Assume constant pressure.

- a A sample of oxygen has a volume of 3.50 mL at 20°C . What volume would it occupy at 80°C ?
- b If a sample of gas had a volume of 1.7 L at 25°C and 1 atm pressure, what:
 - i would its volume be at 100°C ?
 - ii temperature would it need to be cooled to have a volume of 0.80 L?
- c A gas occupies a volume of 60.0 mL at 36°C . What volume would the gas occupy at 0°C ?

Combined gas law

When dealing with gases, we usually find that temperature and pressure change at the same time. By combining Boyle's law and Charles' law, a new relationship called the **combined gas law** is produced. This gives a relationship between volume, pressure and absolute temperature for a given mass of gas. It is most useful in the following form:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

where P_1 , V_1 and T_1 represent initial pressure, volume and temperature (K), and P_2 , V_2 and T_2 represent final pressure, volume and temperature (K).

The temperature must be in measure in kelvin and the units for volume and pressure must be consistent.



GAS LAWS

Use these simulations of Boyle's and Charles' laws.

WOW

Boiling eggs

When a hen lays an egg, the egg is warm. As it cools, the insides contract and a pocket of air is formed. Over time, the porous shell allows carbon dioxide and water to diffuse out and air to enter to equalise the pressure. When the egg is placed in rapidly boiling water, the air quickly expands and causes the shell to crack.

WORKED EXAMPLE 8.3

A sample of gas at a pressure of 105 kPa has a volume of 2.50 L at a temperature of 25°C. What pressure is needed to compress it to 1.0 L at 20°C?

Answer

The pressure would be 258 kPa.

Logic

- 1 Write the relationship:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

- 2 Determine the quantities and convert temperature to kelvin by adding 273:

$$P_1 = 105 \text{ kPa}, P_2 = ?$$

$$V_1 = 2.50 \text{ L}, V_2 = 1.0 \text{ L}$$

$$T_1 = 25^\circ\text{C} = 298 \text{ K}, T_2 = 20^\circ\text{C} = 293 \text{ K}$$

- 3 Check the units are consistent.

- 4 Substitute for values in the formula:

$$\frac{105 \times 2.50}{298} = \frac{P_2 \times 1.0}{293}$$

$$P_2 = \frac{105 \times 2.50 \times 293}{298 \times 1.0}$$

$$P_2 = 258 \text{ kPa}$$

Try these yourself

- A sample of oxygen at 80°C and 135 kPa pressure has a volume of 202 mL. What will the volume of the sample be at 0°C and 101.3 kPa?
- The volume of 1 mole of gas at 25°C and a pressure of 101.3 kPa is 24.5 L. Calculate the molar volume:
 - at the top of Mt Everest: temperature -30°C, pressure 30 kPa.
 - on the surface of Mars: temperature 27°C, pressure 0.80 kPa.
- A 20.0 L car tyre is inflated to a pressure of 200 kPa at 15°C. After an hour's driving, the tyre temperature has risen to 40°C. If the tyre has expanded to a volume of 24 L, what is the tyre pressure?

QUESTION SET 8.3

Remembering

- 1
 - a What does STP stand for?
 - b What are standard laboratory conditions?
- 2 State:
 - a Boyle's law.
 - b Charles' law.

Understanding

- 3 Explain why car owners need to add more air to their tyres in winter and release air from them in summer.
- 4 A 4 L capacity balloon is two-thirds full. If the kelvin temperature of the gas in the balloon doubles, what will happen?

Applying

- 5
 - a A sample of hydrogen gas has a volume of 6.20 L at a pressure of 1.05 atm. What is the volume of the sample if it is compressed until the pressure is 3.00 atm? (Assume temperature remains constant.)
 - b A motorbike tyre with a volume of 24 L is filled with air to a pressure of 2.3 atm. What volume does this air occupy at 1 atm pressure at the same temperature?
- 6
 - a A sample of argon gas has a volume of 450 mL at 100°C. At what temperature will it have a volume of 200 mL if the pressure does not change?
 - b A sample of gas is heated until its temperature doubles from 25°C to 50°C. If its final volume is 100 mL, what was the original volume if the pressure remained constant?
- 7 A sample of helium occupies a volume of 10.0 L at 25°C and a pressure of 1.2 atm. What volume will it occupy at STP?
- 8 The temperature of a closed vessel containing air at a pressure of 103 kPa is raised from 15°C to 200°C. What is the pressure inside the vessel if its volume increases by 10% as the temperature is raised?

Analysing

- 9 A metal cylinder has a safety valve that opens at a pressure of 100 atm. It is to be filled with nitrogen gas heated to 300°C. What is the maximum pressure to which it can be filled at 25°C?
- 10 When a cyclone forms, the wind rushes in to fill an area of low pressure. A cyclone approaches and the air pressure falls from 101.3 to 79.4 kPa.
 - a If the volume of your living room was 430 m³ initially what volume would the air in the room change to so pressure was equalised inside and outside?
 - b Why is it advised that windows be left slightly ajar as a storm approaches?

8.4 Avogadro's hypothesis and molar volumes

In all the examples considered so far, the amount of gas has been kept constant while temperature, pressure and volume have been varied. Adding or removing gas will affect pressure, volume and temperature. Pumping more air into a bicycle tyre will increase the pressure within the tyre. Releasing gas from an inflated balloon causes the volume to decrease and releasing gas from an inflated tyre causes the valve to become quite cold.

In 1811, Amadeo Avogadro attempted to show how the volume of a gas was related to the amount of gas present. He proposed that equal volumes of gas contain equal numbers of particles at the same temperature. This became known as **Avogadro's hypothesis**.

Avogadro's hypothesis states that equal volumes of any gas, measured at the same temperature and pressure, contain the same number of particles.

This means that no matter what the gas is, the total number of particles would be the same under the same conditions. So, provided temperature and pressure are the same, 1 L of hydrogen has the same number of particles as 1 L of helium, or 1 L of oxygen and so on. Also, if the number of particles were doubled, then the volume would double.

Molar volume

A mole is defined as a fixed number of particles (Avogadro's constant, 6.02×10^{23}), so it follows that equal volumes of different gases, measured at the same temperature and pressure contain the same number of moles. This is called the **molar volume of a gas**.

- At standard temperature and pressure (0°C and 100 kPa) the molar volume of a gas is 22.71 L.
- At standard laboratory conditions (25°C and 100 kPa) the molar volume of a gas is 24.79 L.

The relationship between number of moles and volume of a gas can be expressed as:

$$\text{Number of moles of gas} = \frac{\text{volume of gas at STP}}{\text{volume of one mole of gas at STP}}$$
$$n = \frac{V}{22.71}$$

where n is the number of moles and V is the volume of gas (L) at STP.



MORE GAS LAWS

Visit this website for more practice problems using the gas laws

You learnt about the Avogadro constant in Chapter 5.

WOW

Icy cylinders

Condensation and even ice can form on the outside of a gas cylinder as it is being used. As the gas is used and the pressure in the cylinder decreases, stored liquid petroleum gas (LPG) 'boils' to a gas vapour, absorbing heat energy from the steel walls of the container

WORKED EXAMPLE 8.4

- a** How many moles of carbon dioxide gas are present in 350 mL of gas at STP?
b What is the mass of carbon dioxide in part **a**?
c What volume is occupied by 160 g of helium at STP?

Answers

a There is 0.0154 mol.

b The mass is 0.678 g.

c The volume is 908 L.

Logic

1 Write the relationship.

$$n = \frac{V}{22.71} \text{ mol}$$

2 Determine the quantities and check the units.

$$n = ?, V = 350 \text{ mL} = 0.350 \text{ L}$$

3 Substitute in the formula.

$$n = \frac{0.350}{22.71}$$

$$n = 0.0154 \text{ mol}$$

1 Write the relationship.

$$n = \frac{m}{M}$$

2 Determine the quantities.

$$n = 0.0154 \text{ mol}$$

$$M(\text{CO}_2) = 12 + 2 \times 16 = 44 \text{ g mol}^{-1}$$

3 Substitute:

$$0.0154 = \frac{m}{44}$$

$$m = 0.0154 \times 44 = 0.678 \text{ g}$$

1 Write the relationship.

$$n = \frac{V}{22.71} \text{ mol}$$

2 Determine the quantities and check the units.

$$n = ?, V = ?, m = 160 \text{ g}$$

3 Calculate the number of moles.

$$n = \frac{m}{M}$$

$$m = 160 \text{ g}, M(\text{He}) = 4.00 \text{ g mol}^{-1}$$

$$n = \frac{160}{4.00} = 40 \text{ mol}$$

4 Calculate the volume.

$$n = \frac{V}{22.71}$$

$$40 = \frac{V}{22.71}$$

$$V = 40 \times 22.71 = 908 \text{ L}$$



Try these yourself

- a What volume container would you need to store 0.095 mol of nitrogen at STP?
- b
 - i How many moles are present in 0.450 L of oxygen gas at STP?
 - ii What is the mass of oxygen in part i?
- c
 - i What volume will 10 g of hydrogen gas occupy at STP?
 - ii What volume will 10 g carbon dioxide occupy at STP?
 - iii Why are these volumes different when the mass is the same?

The general gas equation

All the gas laws studied so far have required that at least one of the conditions of pressure, temperature, volume and quantity of gas remain constant. In reality, when dealing with gases in situations such as scuba diving and using anaesthetics, all the conditions can vary. However, if Boyle's and Charles's laws are combined with Avogadro's hypothesis, the resulting relationship, called the **general gas equation**, includes all conditions as variables.

The general gas equation is also known as the ideal gas equation and the universal gas equation. It provides a relationship between pressure, volume, temperature and the amount of gas:

$$PV = nRT$$

where P is pressure in kPa

V is volume in L

n is total number of moles of gas present

R is the **universal gas constant** = $8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

T is temperature in K.

The general gas equation, $PV = nRT$, gives the relationship between pressure, temperature, volume and number of moles of a gas.

Regardless of the chemical formula and properties of different gases, the general gas equation describes the physical properties of all gases. The equation is called the *ideal* gas equation because real gases deviate slightly from it, particularly at high pressures and low temperatures, because the intermolecular forces between real gases come into play as the particles slow down and move closer together.

The value of R , the universal gas constant, is dependent on units. The value given above is based on an SI unit for pressure. Sometimes, it is more convenient to use pressure in atmospheres. When the unit for pressure is atmospheres, the value of R is $0.0820 \text{ L atm K}^{-1} \text{ mol}^{-1}$. Table C8.3 shows the units and values for R .

Table C8.3 Units and values for the universal gas constant, R

Condition	$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$	$R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$
Pressure	kPa	atm
Volume	L	L
Temperature	K	K
Amount of gas	mole	mole

It is important to use the correct value of R in the general gas equation. One way of reducing mistakes is to always include the unit and the number when recording the value of a physical quantity.

WORKED EXAMPLE 8.5

What is the mass of carbon dioxide in a sample that occupies 1.50 L at 20°C and 125 kPa?

Answer

The mass is 3.39 g.

Logic

- 1 Write the relationship.

$$PV = nRT$$

- 2 Determine the quantities and check the units.

$$m = ?, P = 125 \text{ kPa}, V = 1.50 \text{ L}, n = ?, T = 20^\circ\text{C} = 293 \text{ K}$$

- 3 Decide on the appropriate value for R based on the units for P .

$$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

- 4 Substitute in the equation.

$$125 \times 1.50 = n \times 8.314 \times 293$$

$$n = \frac{125 \times 1.50}{8.314 \times 293} \\ = 0.077 \text{ mol}$$

- 5 Calculate mass from moles.

$$n = \frac{m}{M}$$

$$M(\text{CO}_2) = 44 \text{ g}$$

$$0.077 = \frac{m}{44}$$

$$m = 0.077 \times 44 = 3.39 \text{ g}$$

Try these yourself

- a What is the volume of 0.34 mol of H_2S gas at 100°C and 75.0 kPa pressure?
- b A sample of oxygen gas is collected and found to occupy a volume of 527 mL at 1.05 atm and 20°C. What is the mass of the sample?
- c What is the pressure of 4.00 g of He in a container of volume 7.5 L at 30°C?

EXPERIMENT 8.2

MEASURING THE MOLAR MASS OF BUTANE

Most disposable cigarette lighters are filled with butane (C_4H_{10}). Butane exists as a gas at room temperature (boiling point, -0.5°C), but is compressed to a liquid in lighters. When the pressure is released (the lighter valve is opened), the butane vaporises and is ignited by a flint in the lighter.

In this experiment, you will determine the molar mass of butane experimentally and then compare this with the theoretical value. The mass of liquid butane released from the lighter will be the same as the mass of gas produced, but the volume will be different.

Aim

To determine the molar mass of butane experimentally and compare this value with the theoretical value

Materials

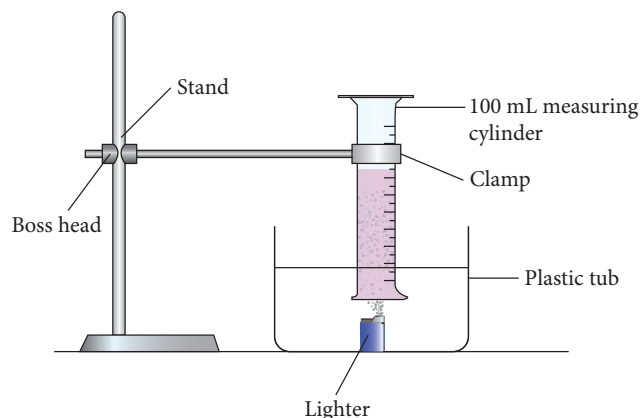
- Pneumatic trough or large plastic tub
- Disposable butane lighter with flint removed
- Thermometer
- 250 mL measuring cylinder
- Retort stand, clamp and boss head
- Barometer
- Water
- Electronic balance
- Paper towel

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Butane is highly flammable.	Ensure the flint is removed from the lighter and there are no open flames in the laboratory.
Butane gas can be irritating.	Work in a well-ventilated area and avoid breathing the gas.

In your write-up, add any more risks you can think of, as well as ways to manage them. In particular, expand on the two risks listed to identify specific risks involved with each of them.

Procedure

- Two-thirds fill the pneumatic trough or large plastic tub with water.
- Wipe the lighter clean and dry, weigh it and record the mass.
- Fill the 100 mL measuring cylinder with water.
- Submerge the cylinder in the tub and gradually turn it upside down.
- Ensure no air bubbles are present. If bubbles are present, then repeat step 4.
- Lift the measuring cylinder just high enough above the base of the tub so the lighter fits underneath. Clamp the cylinder in position (see Figure C8.14).
- Hold the lighter directly under the measuring cylinder and press the lever on the lighter. Ensure all the gas is bubbling into the cylinder.
- Continue to hold down the lever until the water levels inside and outside the measuring cylinder are about the same. Remove the lighter.
- Adjust the height of the measuring cylinder so the water levels inside and outside are the same. Record the volume of gas in the measuring cylinder.
- Thoroughly dry the lighter with paper towel. Be careful not to press the lever.
- Reweigh the lighter and record the new mass.
- Measure and record the temperature of the water in the tub.
- Measure and record the barometric pressure.
- If time allows, repeat steps 2–11.



▲ Figure C8.14
Experimental set-up

Results

Record your results in a table like the one below.

Initial mass of lighter	
Final mass of lighter	
Volume of gas	
Temperature of water	
Barometric pressure	

Analysing the results

- Calculate the mass of butane released.
- Calculate the number of moles of butane.
- Use Table C8.4 on page 360 to help determine the vapour pressure of water at the experimental temperature.
- Calculate the pressure of butane by subtracting the vapour pressure of water from the atmospheric pressure reading.
- Calculate the experimental molar mass of butane.

Table C8.4 Vapour pressure of water at various temperatures

Temperature (°C)	Pressure (atm)	Temperature (°C)	Pressure (atm)
18	0.0204	25	0.0312
19	0.0217	26	0.0332
20	0.0230	27	0.0351
21	0.0245	28	0.0372
22	0.0261	29	0.0395
23	0.0276	30	0.0419
24	0.0295	31	0.0443

Discussion

- 1 Compare the experimental molar mass with the reference molar mass.
- 2 Identify possible sources of experimental error and describe how these would affect the result.
- 3 Why did the vapour pressure of water have to be subtracted to determine the pressure of butane?
- 4 Suggest any improvements you might make to the experiment to make it more accurate.

QUESTION SET 8.4

Remembering

- 1 State Avogadro's hypothesis.
- 2
 - a Define 'molar volume'.
 - b State the general gas equation.

Understanding

- 3
 - a If R is the general gas constant, explain why it has different values.
 - b Why can't the general gas equation be applied at high pressure and low temperature?

Applying

- 4 What volume would a mole of gas occupy at 500°C and 0.80 atm?
- 5 A 5.0 L sample of gas exerts a pressure of 20 kPa at a temperature of 300 K. How many moles of gas are present?
- 6 Calculate the pressure exerted by 2.59 g of carbon monoxide in an 8.0 L flask at 30°C.
- 7 What volume is occupied by 0.200 mol of oxygen gas at 20°C and a pressure of 0.97 atm?
- 8 At STP, the mass of 1.00 L of a gas is 1.89 g. What mass of gas will occupy 1.00 L at 200°C and a pressure of 1.25 atm?
- 9 A sample of noble gas had a mass of 0.20 g and exerted a pressure of 0.98 atm in a container of volume 0.13 L at 27°C. Which noble gas is it?

Analysing

- 10 You have been given three 5.0 L balloons. One is filled with hydrogen, one with carbon dioxide and one with air. The average molar mass of air is 29.0 g mol⁻¹ and the temperature and pressure of all balloons are 27°C and 1.0 atm. Calculate the mass of gas in each balloon and use this to justify why a balloon filled with hydrogen rises while one filled with carbon dioxide sinks.

8.5 Calculating gases in chemical reactions

In Chapter 5, mass calculations based on amounts of reactants and products were considered. The calculations needed to use the mole relationships given by coefficients in a balanced chemical equation in order to determine amounts of reactants and products. This relationship can now be extended to include volumes of gases.

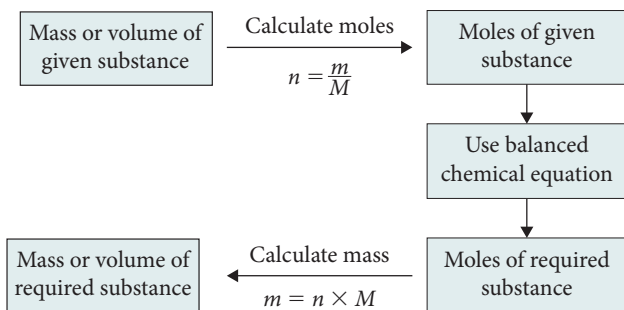


Figure C8.15
Mass-volume relationships, where m is mass in grams, n is number of moles and M is molar mass



GENERAL GAS LAW

Visit this website to practise more problems applying the general gas law.



PVT

Use this simulation to investigate the relationships between pressure, volume and temperature.

WORKED EXAMPLE 8.6

- a** What volume of hydrogen gas would be produced at STP when 0.325 g of zinc reacts with dilute hydrochloric acid?
- b** Sulfuric acid is formed when sulfur dioxide and water react with oxygen in the presence of a catalyst. If 3.20 g of SO_2 react, what volume of O_2 at 200 kPa and 50°C is necessary for complete reaction?

Answers

- a** The volume is 113 mL.

Logic

- 1** Write a balanced equation.



- 2** Calculate the number of moles of Zn using the relationship $n = m/M$.

$$m = 0.325 \text{ g}, n = ?, M(\text{Zn}) = 65.4 \text{ g mol}^{-1}$$

$$n = 0.325/65.4 = 4.97 \times 10^{-3} \text{ mol}$$

- 3** Use the balanced equation to calculate the number of moles of H_2 .

From the equation, 1 mol of Zn produces 1 mol of H_2

$$n(\text{H}_2) = 4.97 \times 10^{-3} \text{ mol}$$

- 4** Use molar volume at STP to calculate the volume of H_2 .

$$n = \frac{V}{22.71}$$

$$4.97 \times 10^{-3} = \frac{V}{22.71}$$

$$V = 4.97 \times 10^{-3} \times 22.71 = 0.113 \text{ L} = 113 \text{ mL}$$

- b** The volume is 0.336 L

- 1** Write a balanced equation.



- 2** Calculate the number of moles SO_2 using the relationship $n = m/M$.

$$m = 3.20 \text{ g}, n = ?, M \text{ can be calculated}$$

$$M = 32.1 + 2 \times 16 = 64.1 \text{ g mol}^{-1}$$

$$n = \frac{3.20}{64.1} = 0.0499 \text{ mol}$$

- 3 Use the balanced chemical equation to calculate the number of moles of O_2 and H_2SO_4 .

From the equation, 2 moles of SO_2 reacts with 1 mole of O_2 .

$$n(O_2) = \frac{1}{2} \times n(SO_2)$$

$$n(O_2) = \frac{1}{2} \times 0.0499 = 0.025 \text{ mol of } O_2$$

- 4 Determine the volume using $PV = nRT$.

$$P = 200 \text{ kPa}, V = ?, n = 0.025 \text{ mol}, T = 50^\circ\text{C} = 323 \text{ K},$$

$$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$200 \times V = 0.025 \times 8.314 \times 323$$

$$V = \frac{0.025 \times 8.314 \times 323}{200} = 0.336 \text{ L}$$

Try these yourself

- a A mass of 4.86 g of magnesium is ignited in nitrogen dioxide. The products of the reaction are magnesium oxide and nitrogen gas. Calculate the volume of nitrogen gas produced at 273 K and 101.3 kPa.
- b The complete combustion of carbon in oxygen produces carbon dioxide. Calculate the volume of oxygen that would react at SLC with 10.0 g of carbon and the volume of carbon dioxide produced under the same conditions.
- c What mass of potassium chlorate must be heated to give 120 mL of oxygen at 1.3 atm and 32°C ?
The equation for the reaction is:
 $2\text{KClO}_3(\text{s}) \rightarrow 2\text{KCl}(\text{s}) + 3\text{O}_2(\text{g})$
- d Hydrogen peroxide (H_2O_2) decomposes to give water and oxygen gas. How many grams of hydrogen peroxide are needed to produce 100 L of oxygen at SLC?
- e 10.01 g of calcium carbonate is decomposed by heat to produce calcium oxide solid and carbon dioxide gas. What volume of carbon dioxide gas is formed at 20°C and 110 kPa?
-

CHAPTER CHECKLIST

You should know:

- gas pressure is caused by collisions of gas molecules with the walls of a container
- the main units of gas pressure are pascal (Pa) and atmosphere (atm)
- the kinetic theory of gases explains the physical properties of gases in terms of the motion of their molecules and proposes that:
 - 1 gases consist of molecules (except the noble gases, which consist of atoms) that move in continual random straight-line motion
 - 2 the average distance between gas molecules is very large compared to the size of the molecules
 - 3 intermolecular forces between molecules are negligible
 - 4 all collisions of gas molecules are perfectly elastic, which means no net energy loss occurs
 - 5 pressure is due to collisions of the molecules with the walls of the container
 - 6 temperature is a measure of the average kinetic energy of the molecules
- absolute temperature is measured in kelvin according to the absolute temperature scale, which has the same-sized divisions as the Celsius scale but an end point of zero, called absolute zero
- Boyle's law states that at a constant temperature, the volume of a given mass of gas is inversely proportional to the applied pressure
- Charles' law states that at constant pressure, the volume of a fixed quantity of gas is proportional to its absolute (kelvin) temperature
- the combined gas law states a relationship between initial and final volume, pressure and absolute temperature for a given volume of gas
- Avogadro's hypothesis states that equal volumes of any gas, measured at the same temperature and pressure, contain the same number of particles
- the general gas equation, $PV = nRT$, gives the relationship between pressure, temperature, volume and number of moles of a gas.

You should be able to:

- convert between absolute and Celsius temperature scales
- select and use appropriate mathematical representations to solve problems, including using the mole concept and general gas equation to calculate the mass of chemicals and/or volume of a gas (at standard temperature and pressure) involved in a chemical reaction.

CHAPTER GLOSSARY

absolute temperature temperature measured in kelvin according to the absolute temperature scale, which has the same sized divisions as the Celsius scale but an end point of zero

absolute zero lowest possible temperature (0 K or -273.15°C)

atmospheric pressure pressure due to the atmosphere; at sea level has an average value of 1 atm

Avogadro's hypothesis equal volumes of any gas, measured at the same temperature and pressure, contain the same number of particles

Boyle's law a constant temperature, the volume of a given mass of gas is inversely proportional to the applied pressure

Charles' law at constant pressure, the volume of a fixed quantity of gas is proportional to its absolute (kelvin) temperature

combined gas law a relationship between initial and final volume, pressure and absolute temperature for a given mass of gas

diffuse the spreading out of a gas to fill a space

elastic collision a collision in which there is no net loss of energy

general gas equation an equation that gives the relationship between pressure, temperature, volume and number of moles of a gas as $PV = nRT$

ideal gas a gas that perfectly obeys all the proposals of the kinetic theory of gases and the gas laws

kinetic theory of gases a theory that explains the physical properties of gases in terms of the motion of their molecules

molar volume of a gas the volume occupied by 1 mole of a gas at known temperature and pressure

pascal (Pa) the SI unit of gas pressure, equivalent to a force of 1 newton per square metre

pressure the force exerted by a gas or gases due to the collisions of the particles with the walls of a container; expressed as force per unit area

universal gas constant (R) a constant value used in the general gas equation, equal to $8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ or $0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$

CHAPTER REVIEW QUESTIONS

Remembering

- 1 **a** What is the difference between a real gas and an ideal gas?
b When does this difference become significant?
- 2 To convert °C to K you:
A subtract 273.
B add 373.
C add 273.
D it depends on the pressure.
- 3 Which of the following best expresses Boyle's law?
A $V \propto \frac{1}{P}$
B $T \propto V$
C $P \propto V$
D $V \propto \frac{1}{T}$
- 4 Gases are more easily compressed than liquids and solids. The reason for this behaviour is that:
A gas molecules move with much greater velocities than the molecules of liquids and solids, permitting gases to adjust more rapidly to a change in volume.
B gas molecules undergo elastic collisions with the walls of a container and elastic substances are easily compressed.
C the average distance between gas molecules is much greater than that between particles in liquids or solids, so the volume may be reduced more easily.
D attractive forces between gas molecules are much smaller than those between particles in solids and liquids, and these small forces can be readily overcome during compression.
- 5 Which of the following is *not* one of the assumptions of the kinetic theory of gases?
A All collisions of gas molecules are non-elastic. This means energy is lost during collisions.
B Gases consist of tiny particles called molecules, except for noble gases, which consist of atoms.
C The molecules of a gas exert negligible attractive or repulsive forces on each other.
D The average distance between molecules in a gas is very large compared with the size of each gas molecule.

Understanding

- 6 A gas fills a 100 mL cylinder fitted with a piston at a particular temperature and pressure. If the volume of the gas is halved by pushing in the piston, and at the same time the absolute temperature is doubled, the pressure of the gas will:
A increase four times.
B be doubled.
C be unchanged.
D be halved.
- 7 Which of the following statements about the properties of gases is correct?
A The solubility of a gas in a liquid increases as the liquid is heated.
B One mole of any gas at STC occupies 24.5 L.
C Larger gas molecules diffuse more quickly than smaller molecules.
D The thermal energy of a gas is given by its temperature.
- 8 Decide whether each of the following statements is true or false. For those that are incorrect, write the correct statement.
a Halving the number of particles in a given volume of gas decreases the pressure by one-half if the temperature is kept constant.
b At absolute zero, the volume occupied by a gas would be zero.

c At constant temperature and pressure, the volume of a gas decreases as the number of moles of the gas increase.

d According to the combined gas law, $T_2 = P_1 \times \frac{V_2 \times T_1}{P_2 \times V_1}$.

- 9 A sample of methane has a volume of 4.60 L at 27°C. What volume would it occupy at 120°C if the pressure remained constant?

Applying

- 10 As a man dives into the sea, the pressure of gas in his lungs changes from 100 000 Pa to 150 000 Pa. If his lungs initially held 6 L of gas, the volume of his lungs would become (assume temperature remains constant):

A 3 L.

B 4 L.

C 6 L.

D 9 L.

- 11 Ammonia was prepared by the following reaction:



If 1 L of nitrogen is reacted with excess hydrogen, how many litres of ammonia will be produced?

A 2 L

B 3 L

C 1 L

D 0.5 L

- 12 What is the final pressure of gas (at a constant temperature) if 50 mL at 100 kPa expands to 60 mL?

- 13 An industrial plant manufacturing polythene (polyethylene) stores ethene (ethylene) gas (C_2H_4) in a vessel with a capacity of 2 L and capable of withstanding a pressure of 1500 kPa. What is the maximum mass of ethene that could be stored in the vessel at a temperature of 42°C?

- 14 350.0 mL of a gas at 26°C and 105.6 kPa has a mass of 0.35 g. Calculate the molar mass of the gas.

- 15 Carbon monoxide burns in oxygen according to the equation:



Calculate the volume of oxygen required at 25°C and 1.00 atm for the combustion of 28 g of carbon monoxide.

- 16 Ammonia gas may be prepared in the laboratory by heating a paste of ammonium chloride and calcium hydroxide according to the reaction:



If 107 g of ammonium chloride reacted fully with excess $\text{Ca}(\text{OH})_2$, what volume of ammonia is produced at 0°C and 1.0 atm pressure?

Analysing

- 17 An experiment was performed to determine the effect of pressure on the volume of a fixed mass of ammonia at 25°C. The following readings were obtained.

P (atm)	0.1000	0.2000	0.4000	2.000	8.000
V (L)	244.5	122.2	61.02	12.17	2.925

Plot these results on graph paper.

Ammonia deviates from an ideal gas at higher pressures. On the same graph, plot the expected volume for all the pressures. (Assume the first pair of readings are correct.)

18 Nitroglycerine has a density of 1.59 g mL^{-1} . It explodes to form several gases according to the reaction:



A sealed 1.00 mL container is filled with nitroglycerine and detonated. Assuming standard temperature and that the container would not break upon detonation, calculate the pressure inside the container in atmosphere.

Reflecting

19 A car with typical petrol consumption of about 14 km L^{-1} will produce about 0.16 kg of carbon dioxide per kilometre.

a If the typical annual distance travelled is 20 000 km, how much carbon dioxide will it produce in a year?

b There are about 1 billion (10^9) motor vehicles in the world. How many kilograms of carbon dioxide do they produce per year?

This is a lot of carbon dioxide but how does it compare with the amount of CO_2 in the atmosphere? The volume of CO_2 in the atmosphere is approximately $1.8 \times 10^{18} \text{ L}$.

c Assuming the temperature is 0°C , calculate the mass of CO_2 in the atmosphere.

d Using your answer from part **b**, calculate the fraction of CO_2 cars contribute to the atmosphere.

e If a fully grown tree can convert about 300 kg of CO_2 a year to oxygen, how many trees would there need to be to offset the effects of motor vehicle emissions of CO_2 ?